PHYS4010 Laser use Safety Training (SPRING 2025)

Introduction:
You will be using a Class 3 b Helium Neon Laser with a Continuous Wave (CW) output of between 10 and 30 mW at a wavelength of 632.8 nm. We may also use laser pointers. Typical pointer lasers with outputs of less than 5.00mW and wavelengths of ~650nm (red) or 532nm (green) are typically labeled as Class 2. The Class 3 b laser rating has an upper power range of 500mW—so we are using lasers at the lower end of the class 3 b power scale. With care these are relatively safe to use—with precautions.

Because we are in an academic, rather than medical or industrial setting we do not have enclosures for laser beams (enclosures typically reclassify all systems to Class 1 due to limited exposure). We will need access to the laser beam and will be moving reflective optics. The procedures for safe use and general training issues are outlined here.

Safety procedures are always being updated to improve safety and handling of laser light. However, students have safely used these lasers without incident for 25 years (probably longer but that is as long as my institutional memory goes). Note that beam divergence, care on setup, natural motion of eye, typical motion of reflection sources (during alignment), and care taken not to have direct FOCUSED specular (mirror like) reflections stationary in your face—accomplish most of what is needed to ensure a danger free environment. For the powers and laser conditions we use---harmful incidents are unlikely.

If you subjected your eye to a FULL beam reflection (unlikely) with very little beam divergence (unlikely—requires use of no additional optics and placing face in laser beam), and moved a reflecting item so that the beam sweeps through about 30^{o} in one tenth of a second (fairly slow motion)-----then you would be approaching maximum permissible exposure levels for our lasers, beam diameters, divergences and powers. Note that uncoated glass reflects about 4% of incident light. With reasonable care you will never encounter an eye hazard with these lasers.

Tasks.

1. Download a copy of the current updated Augusta University Laser Safety Guide Draft from my website (probably linked next to this document). Read it.
2. Watch the laser safety training videos (linked same place) complete this. ------
    a. I am looking into the updated links on these (from Flash to ??? something modern).
    b. I'll notify you if these get updated.

3. Complete the laser safety training form after we have discussed laser safety in class (first day). Sign and hand in.

In class/lab procedures for laser use (Spring 2023 Standard Operating Procedure for lasers).

1. We have HeNe power supplies connected to each laser. The keys should be in the off positions. Do not leave rooms unattended or unlocked when lasers are out.
2. When disconnecting the power supply from the laser tube—MAKE SURE THE LASER IS TURNED OFF FIRST. OTHERWISE YOU MAY DESTROY THE LASER.
3. When using the laser the door to the room 3063 should be closed and no untrained (unauthorized) persons should be in the room.
4. The "Laser in use" light should be switched on.
5. During alignment procedures you should take care to keep eyes out of beam paths.
   a. You should move optics slowly and deliberately
   b. Think about where reflections may occur
   c. Make sure you do not endanger your lab partner, others in the room.
   d. Let others know when you are performing alignments that may send light their way.
6. We have HeNe laser blocking goggles available—however---since this light is visible, the goggles also obstruct important viewing (visible light). You may choose to wear these upon request, but take care to take them on and off for proper viewing. After alignment you may wear HeNe laser blocking goggles for additional safety. You should not use these goggles if you need to do continued viewing or alignment of HeNe light shining upon a wall or other observation surface.
7. No visitors (you are responsible for controlling the room during laser use).
8. Remove all jewelry, watches, etc from necks, wrists, arms or other possible reflecting locations. Be on the lookout for any stray reflections.
9. Ensure that the laser beam path is clear of extraneous reflecting obstructions.
10. Place and use appropriate beam stops (stable boxes or other diffusely reflecting surfaces suffice). You may notice stray reflections---while avoidable, and probably not exceeding maximum exposure limits, boxes or other stops serve as excellent movable beam stops. I use them frequently.
11. Face away from any beam which might be pointing in your direction (you and partner) when turning laser on.
    a. When you are taking data and viewing scattered or reflected laser light---DON'T ---DO NOT---DO NOT ---DON'T TURN AROUND TO STARE INTO THE LIGHT.
    b. It takes deliberate care to how you move so that you don't turn around and stare into the light. Be careful, be deliberate.
12. Note any reflections ---move so that your body does not obstruct.
13. Block any additional reflections which may cause a hazard to you or others (look around the room).
14. If you leave the room, make sure the laser is off—or supervised---or moved back to it's "home position".
15. Chairs and sitting are not permitted in this room to ensure that eyes are not at the same level as the laser beam. There are a few chairs that can be used in front of a computer for brief times. In general, stand!
16. Since we use open beam paths and perform alignment you are to treat the entire room as a Nominal Hazard Zone when lasers are in use (watch out for any stray reflections you may have missed).
17. Fill out and hand in Laser Safety Training Completion form prior to using laser.

Laser Safety Training Completion
PHYS4010 Authorized Users of Class 3 b Helium Neon Lasers in room W3002 Science Hall
Spring 2025

1. I ____sign then print, have reviewed all laser training materials provided by Dr. Colbert (spring 2025) and understand my responsibilities regarding safe use of Class 3 b HeNe Lasers during this course. My responsibility ensures both my own safety and that of others.

2. I have downloaded and read the Augusta University Laser Safety Guide Draft statement.

3. I have viewed the required online videos required for laser safety training. (If Available)

4. I agree to abide by the standards Dr. Colbert has set forth for operation of lasers—in this document.